

Continuous Prob

$$i) \int_{-\infty}^{\infty} f(x) dx = 1$$

ii) $f(x) \geq 0$ (non-neg func over range)

$$iii) P(0 < x < b) = \int_0^b f(x) dx$$

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx \quad \text{cumulative}$$

3.11

Suppose error in the reaction temp in degrees °C for controlled laboratory experiment is a cont random var x having prob density func

$$f(x) = \begin{cases} x^2/3 & -1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

a) verify it is proper Density func.

$$f(x) \geq 0 \quad \int_{-\infty}^{\infty} f(x) dx = \int_{-1}^2 \frac{x^2}{3} dx = \frac{x^3}{9} \Big|_{-1}^2 = 1$$

b) Find prob using PDF

$$P(0 < x < 1) = \int_0^1 f(x) dx$$

$$= \int_0^1 \frac{x^2}{3} dx = \frac{x^3}{9} \Big|_0^1 = \frac{1}{9}$$

c) using CDF

$$F(x) = P(X \leq x)$$

$$= \int_{-1}^x \frac{x^2}{3} dx = \frac{x^3}{3} \Big|_{-1}^x$$

$$= \frac{x^3}{3} + \frac{1}{3} = \frac{x^3 + 1}{3}$$

Prob from 0 to 1

$$= F(1) - F(0)$$

$$\left[\frac{(1)^3 + 1}{3} \right] - \left[\frac{(0)^3 + 1}{3} \right]$$

$$= \frac{1}{3}$$

Note

From CDF to PDF
take der of $F(x)$.

Mathematical Expectation and Variance.

4.10

$$f(x) = \begin{cases} 2(x-1) & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

mean and var of x

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$= \int_1^2 x \cdot 2(x-1) dx$$

$$= 2 \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_1^2 = \frac{5}{3}$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$= \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$= \int_1^2 x^2 \cdot 2(x-1) dx$$

$$= 2 \left[\frac{x^4}{4} - \frac{x^3}{3} \right]_1^2 = \frac{17}{6}$$

$$\text{Var} = \frac{17}{6} - \left(\frac{5}{3} \right)^2$$

$$= \frac{1}{18}$$

$$Q) f(x) = \begin{cases} x^3/3 & -1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

And expectation of $4x+3$.

$$g(x) = 4x + 3$$

$$\begin{aligned} E[g(x)] &= \int_{-\infty}^{\infty} g(x) f(x) dx \\ &= \int_{-1}^2 (4x+3) \frac{x^2}{3} dx \end{aligned}$$

$$= 8$$

$$\text{Var } g(x) = E[g(x)^2] - E[g(x)]^2$$

$$= \int_{-1}^2 (4x+3)^2 \frac{x^2}{3} dx$$

$$= \frac{371}{5}$$

$$\text{Var} = \frac{51}{5}$$

Q1) Total hours measure in unit of hundred hours that a family runs a vacuum cleaner over one year has density func.

$$f(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Find Prob that over one year a family runs it

a) $x < 120$

$P(X < 120)$

$$\begin{aligned} P(X < 1.2) &= \int_0^1 f(x) dx + \int_1^{1.2} f(x) dx \\ &= \int_0^1 x dx + \int_1^{1.2} (2-x) dx \\ &= \frac{x^2}{2} \Big|_0^1 \left[2x \Big|_1^{1.2} - \frac{x^2}{2} \Big|_1^{1.2} \right] \end{aligned}$$

$= 0.68$

b) $P(0.5 < x < 1)$

$$\int_{0.5}^1 x dx$$

$= 0.375$

Prob 4.12, 20, 29, 50, 101.